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ITAI 1371: Intro to Machine Learning

September 8, 2026

L04 Reflection Journal

Introduction

The purpose of this lab was to understand Exploratory Data Analysis, using a dataset on Titanic. I examined the characteristics of the relationship between variables, missing values and possible anomalies within the dataset.

Reflection 1: The port of embarkation finding

I created a plot to show the different ports passengers boarded from. My plot showed Cherbourg as the only port where more passengers survived than died. Rather than take it at face value, I created a temporary cross tab checking what else differed between ports and found Cherbourg boarded significantly more first-class passengers.

Reflection 2: Fare and what a plot revealed beyond numbers

The boxplot looked incorrect at first. Both boxes were squashed at the bottom with outliers trailing. But after using the boxplot, I realized the finding was the shape. I considered a log scale but stayed with the standard boxplot, since cropping the outliers would have hidden real data. Fare is heavily skewed, and a mean would most likely have hidden the skew, the clustering or the trailing outliers.

Reflection 3: Connection to workflow

The phases are as follows: data collection and understanding, data preparation, then model training and evaluation. EDA comes before modeling and shapes the next steps. You examine what you have before choosing an algorithm, because what the data looks like determines the most appropriate model.

Conclusion

Insights can be revealed through shapes, numbers, relationships, or lack thereof. Rule of thumb: Do not accept data at face value. The port of embarkation looked like it mattered on its own until I checked what else differed between ports. The fare boxplot looked broken until I realized the shape was the finding. In both cases, questioning the pattern rather than accepting the data the way it was helped come to my conclusion.

Diane Lenhoff

Lab 04 Reflection Journal

In this lab, I learned how Exploratory Data Analysis (EDA) helps us understand data before building a machine learning model. Using the Titanic dataset, I was able to see how factors such as gender, passenger class, age, fare, and port of embarkation were related to survival.

What stood out to me the most was that females and first-class passengers had a much higher chance of surviving. I also found it interesting that passengers who survived generally paid higher fares.

The graphs made it much easier for me to see patterns that I might have missed by just looking at numbers. I also learned that just because two things are related does not mean that one caused the other. Overall, this lab helped me better understand how EDA and visualizations can be used to find important patterns in data.

Knowledge Check Instructions: Answer the following questions in this markdown cell.

1. ****What is the primary goal of Exploratory Data Analysis (EDA)?**

The main goal of EDA is to understand the data before building a model. It helps us find patterns, relationships, missing data, and anything unusual in the dataset.

2. Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic? (Describe them in terms of class, gender, and age).

Based on the plots, a female passenger in first class who was younger had a better chance of surviving.

3. Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't?

Visualizing the data makes it easier to see what is really going on. A graph can show patterns, differences, and unusual data that might not be obvious just by looking at averages or counts.

Aaliyah Phillips

ITAI 1371

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Module 4 Reflective Journal

Going into this module, I had some previous knowledge regarding EDA from my previous course Statistical Programming with R. So, I am already familiar with process of exploring data through plotting and visualization. Using R and using Colab are very similar, though I believe R has a more advanced user interface compared. Plotting on both is pretty much the same, so much of this week's lab was a necessary knowledge refresh for me.

When I loaded up the Titanic dataset, I went over the data and noticed that the "age" and "cabin" columns had missing values. I kept note of this in case it came up again later. Studying the summary statistics, I found it difficult to extract conclusions from the stats alone. For example, deciphering the exact number of people that died or survived was impossible here. However, with the 1st visualization I was able to see the number of people who died surpassed the people who survived. As I continued through all the visuals, I was able to extract the survival rate based on characteristics such as class, gender, age, and fare. A person who was: in 1st class, female, under the age of 40, and paid a higher fare had the highest likelihood of survival.

Moving onto the experiment portion, by creating my own visualizations I was able to see that those who embarked through Cherbourg had the highest chances of survival, while those who embarked on the Southampton were the least likely to survive. On the second experiment, I used a violin plot to see that those who paid a higher fare had higher survival rates. Which falls in line with the insight from earlier that those who were of a higher class were more likely to survive. Out of all the labs so far, this one was the most enlightening. It showed me how much EDA is pivotal to understanding the nuances aspects of data and that just stats alone cannot tell you the whole story.

L04 Reflective Journal – ITAI 1371

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This lab walked through Exploratory Data Analysis (EDA) using the Titanic dataset, and it gave me a much clearer picture of why EDA is considered a foundational step before any modeling work begins. Going into it, I understood EDA conceptually as "looking at your data first," but actually working through the notebook made that idea concrete. Each visualization uncovered a pattern that wasn't obvious from the raw table alone, and it reinforced how much information gets lost if you jump straight into building a model without exploring the dataset's structure and relationships first.

The pre-built visualizations in the lab, survival by class, survival by gender, and the age distribution, were straightforward to follow, but the real learning came in the student experimentation section. For Experiment 1, I built a countplot comparing survival by port of embarkation, which showed that passengers who boarded at Cherbourg had a noticeably higher survival rate than those who boarded at Southampton or Queenstown. At first this seemed unrelated to survival, but connecting it back to the earlier class-based plot made it clearer, Cherbourg had a higher share of first-class passengers, which likely explains the pattern rather than the port itself having any direct effect. That was a useful reminder that variables in a dataset are rarely independent, and a pattern in one plot often has its real explanation sitting in another.

For Experiment 2, I compared fare against survival using a boxplot, since fare is a continuous variable and a boxplot makes it easy to compare medians, spread, and outliers between the two survival groups. Passengers who survived tended to have paid higher fares on average, which lines up with the class and embarkation patterns from earlier. Choosing the boxplot over a violin plot was a small but deliberate decision, I wanted a clean comparison of the central tendency and spread rather than the full shape of the distribution, and the boxplot communicated that more directly.

The biggest takeaway for me was the Knowledge Check question about why visualization matters beyond summary statistics. A single number like a mean or a survival rate can hide a lot, two very different distributions can produce the same average. Seeing the plots made patterns like the class gap and the gender gap immediately obvious in a way that scrolling through `.describe()` output never would have. That distinction matters for where I want to take my career. As I move from cybersecurity into AI security, being able to quickly spot anomalies, imbalances, or suspicious patterns in data visually, rather than only through statistics, is a skill that will carry over directly, whether that's spotting irregular network traffic, biased training data, or unexpected model behavior.

Overall, this lab reinforced that EDA isn't just a preliminary checkbox before modeling, it's where you actually build your understanding of what the data is telling you, and that understanding shapes every decision that comes afterward.

Reflection Journal: EDA on the Titanic Dataset

This lab treated Exploratory Data Analysis (EDA) as investigative detective work, serving as an introduction to formal modeling that helps make hypotheses and expose underlying data structures.

Initial Inspection & Quick Experimentation

Loading data via pandas provided initial clues: `df.info()` and `df.describe()` revealed missing values in 'Age' and 'Cabin' and established a baseline ~38.3% survival rate. Secondary checks like `pd.crosstab` on embarkation ports offered initial insights but required deeper visual analysis.

The Power of Visual EDA vs. Raw Statistics

Tabular summaries often mask critical patterns and non-linear relationships.

Visualizing data with Seaborn and Matplotlib transformed the analysis:

- **Revealing Subgroups:** While the average age was ~29.7, FacetGrid age plots revealed a distinct survival spike among young children that single summary metrics compress.
- **Multivariable Interactions:** Overlaying survival on gender and passenger class in countplots clearly demonstrated protocol adherence ("women and children first") far better than complex pivot tables.

Socioeconomic Status and Survival Odds

Socioeconomic standing had a decisive impact on passenger survival:

- **Class Disparities:** Comparing 'Pclass' to 'Survived' showed a decline in survival probability from first to third class.

- **Fare Distributions:** 'Fare' boxplots confirmed survivors occupied higher price brackets and highlighted extreme outliers, showing how financial resources directly influenced rescue priority.

Conclusion

EDA is the foundation of data projects. Pairing statistical summaries with targeted visualizations enables data scientists to uncover critical structural biases.